## Before AI:

https://docs.google.com/spreadsheets/d/14fpw2edzZV8etOHjjeOs7vJ9F_022TG2ObSlos6V82Q/edit?usp=sharing

## After AI:

https://docs.google.com/spreadsheets/d/1jZmyVAMcSubVG_vam-NdkCbV39tJF0vDgXxnSYnmlF0/edit?usp=sharing

1. Delete derived columns. There are 3NF violations (e.g., OrderItem.subtotal, Order.totalAmount, Section.capacity, Venue.capacity, Waitlist.position).
    a. Decision: Accept
    b. Why? : 3NF Strictness & Anomaly Prevention: A fundamental principle of 3NF is that non-key attributes must not transitively depend on other non-key attributes. Derived attributes (such as subtotal = quantity * unitPrice, totalAmount = subTotal - discountAmount, or sum aggregations like Venue.capacity) introduce redundant storage. If the underlying data changes without updating the derived column, update anomalies and data inconsistency occur.

2. Delete orderID, ticketTypeID and seatID. They are reachable through orderItemID and eventSeatID, so they are redundant.
    a. Decision: Modify
    b. Why? : I agree with linking Ticket to orderItemID, since the order and ticket type can be found through OrderItem and 1 order item can produce several tickets. I modified eventSeatID to be a nullable FK, because the recommendation assumes every ticket has a seat. Our system also sells general admission tickets (TT004 and TT005), which may have no assigned seat. A required eventSeatID would force fake seats. Keeping it unique still stops one seat being sold twice, and the table stays in 3NF.
3. Organizers are also customers of their own events. U001 owns O001 and also buys, waitlists and reviews E001. Add users U007–U010 and point `Organizer.userID` at them.
    a. Decision: Reject
    b. Why? : This is a business rule, not a normalization issue. 3NF only checks functional dependencies and keys. An organizer's userID that also appears in Order.userID or Review.userID breaks neither, so the schema is already valid. Changing the sample users to U007–U010 would only edit example data and would not improve the structure. If the system should stop organizers from buying tickets to their own events, that belongs in application logic or a CHECK-style constraint, not in the normalization.

# Table

| AI Recommendation | Decision | Why? |
|---|---|---|
| Delete derived columns that break 3NF. | Accept | Stored totals can drift out of date when the rows behind them change. Each value can be calculated from other columns, so we compute it instead. |
| Replace Ticket's orderID, ticketTypeID and seatID with orderItemID and eventSeatID. | Modify | Not every ticket has a seat (general admission, TT004 and TT005). We kept orderItemID, but made eventSeatID nullable and unique, so no fake seats are needed and one seat still can't be sold twice. |
| Give organizers their own user accounts (U007–U010), since U001 organizes and buys for E001. | Reject | This is a business rule, not a normalization issue: a shared userID breaks no key or dependency. We enforce it in application logic or a trigger. |